$$D_4 = 20 + \left(\frac{26.8 - 12}{19}\right) \times 10 = 27.79$$

P_{75} = Size of $\left(\frac{75N}{100}\right)^{th}$ value = 50.25^{th} value. Thus P_{75} lies in the class (50 – 60) and its corresponding values are

$$L = 50;\ \frac{75N}{100} = 50.25;\ pcf = 46, f = 9, c = 10.$$

$$P_{75} = L + \left(\frac{\frac{75N}{100} - pcf}{f}\right) \times c$$

$$= 50 + \left(\frac{50.25 - 46}{9}\right) \times 10 = 54.72$$

8.1.4 Geometric mean

Geometric mean is defined as the nth root of the product of *n* observations or values.

If there are two observations, we take the square root; if there are three observations we have to take the cube root and so on

$$GM = \sqrt[n]{X_1 . X_2 . X_3 ..... X_n} = (X_1, X_2, X3, ... X_n)^{1/n}$$

where $X_1, X_2, X_3, ..., X_n$ refer to the various items of the series which are all greater than zero and n refers number of observations.

Thus the geometric mean of 3 values 2,3,4 would be

$$GM = \sqrt[3]{(2)(3)(4)} = 2.885$$

When the number of items is three or more the task of multiplying the numbers and of extracting the root becomes excessively difficult. To simplify calculations, logarithms are used. Geometric mean is calculated as follows:

$$\log GM = \frac{\log X_1 + \log X_2 + ... + \log X_n}{n}$$

$$\text{(or)} \quad \log GM = \left(\frac{\Sigma \log X}{n}\right)$$

$$GM = Anti\log\left(\frac{\Sigma \log X}{n}\right),$$

where n is number of observation.

(i) In discrete observation

$$GM = Anti\log\left(\frac{\Sigma f \log X}{N}\right); \text{ where } N = \Sigma f$$

(ii) In Continuous observation

$$GM = Anti\log\left[\frac{\Sigma f \log m}{N}\right];$$ where m is midpoint and $N = \Sigma f$

Example 8.5

Daily income (in Rs) of ten families of a particular place is given below. Find out GM

85, 70, 15, 75, 500, 8, 45, 250, 40, 36.

Solution:

X	*log X*
85	1.9294
70	1.8451
15	1.1761
75	1.8751
500	2.6990
8	0.9031
45	1.6532
250	2.3979
40	1.6021
36	1.5563
	$\Sigma \log X = 17.6373$

Table : 8.3

$$GM = Anti\log\left(\frac{\Sigma \log X}{n}\right); \text{ where } n = 10$$

$$GM = Anti\log\left(\frac{17.6373}{10}\right)$$

$$= Anti\log(1.7637)$$

$$GM = 58.03$$

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